Oleksandr Obramenko

Senior Frontend & AI-Native Engineer

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PROFESSIONAL SUMMARY

Senior Frontend and AI-Native Engineer with over **8 years** of professional software engineering experience across the modern web development landscape, with strong expertise in **Angular (17–21+), React, Next.js, TypeScript, RxJS, Signals, and NgRx Signal Store**. Experienced across the full software development life cycle—from architectural design and feature implementation to debugging, performance optimization, and building complex, scalable, high-performance applications. Skilled in constructing hybrid micro-frontend architectures, SSR/SSG implementations, and integrating cutting-edge **AI-native workflows, LLMs, Model Context Protocol (MCP), Agentic Workflows, and Multi-Agent Swarms**.

TECHNICAL SKILLS

Core Frontend & Frameworks: React 18/19, Next.js (App Router, Server Components), Angular (v5–v21, Signals, Signal Store, RxJS), TypeScript, Modern JavaScript (ES6+), Micro-Frontends (Nx Monorepo, Module Federation), Core Web Vitals Optimization, HTML5 & CSS3/SCSS, Tailwind CSS, CSS-in-JS (Styled Components, Emotion).

State Management & Data Fetching: TanStack Query (React Query), Zustand, Redux Toolkit (RTK), React Context API, Angular Signal Store, NgRx, Apollo Client (GraphQL), SWR, Jotai, Recoil.

AI-Native Engineering & Agentic Workflows: Model Context Protocol (MCP), Agent Swarm Orchestration, Custom Agent Skills (`SKILL.md`), System Instructions & Prompt Engineering, LLM Strategy & Token Optimization, GitHub Copilot Integration, AI-assisted code refactoring & code reviews.

Testing & Quality Assurance: Unit & Integration Testing (Jest, React Testing Library, Jasmine, Karma), End-to-End (E2E) Testing (Cypress, Protractor), Browser Performance Profiling (React DevTools, Network/Performance Tab, Bundle Analysis).

Architecture, Tools & CI/CD: Front-End Architecture & System Design, Component-Driven Development, Build Tools (Vite, Webpack, Babel), Version Control (Git), CI/CD Pipelines, Azure DevOps, RESTful APIs & GraphQL, Stripe & Google Play Billing Integration.

WORK EXPERIENCE

Insilico — Senior JavaScript & AI-Native Engineer

11/2023 – 08/2025

- Architected and developed complex frontend and hybrid fullstack applications within the biopharmaceutical domain utilizing **Nx monorepo**, Angular 17–21, Angular Signal Store, and embedded **React / Next.js** dashboard modules.
- Engineered custom **MCP (Model Context Protocol)** servers and automated **AI Agent Workflows** using `agents.md`, system instructions, hooks, tool calling, and prompt engineering to streamline data transformation pipelines.
- Integrated **Multi-Agent Swarm Orchestration** and custom LLM **Agent Skills (`SKILL.md`)** to automate UI component scaffolding, PR verification, and unit test generation.
- Managed model selection based on task constraints, optimizing token budgets and execution costs while maintaining strict enterprise code quality standards.
- Engineered dynamic graphical representations and rendered interactive canvas elements using **GoJS**.
- Integrated **Escher** library to build and visualize interactive biological flux maps for domain-specific data workflows.
- Implemented multi-type data visualizations and comparison dashboards using **Apache ECharts** and PrimeNG component library.
- Designed and established a scalable, maintainable frontend architecture tailored to support performance-intensive clinical and domain requirements.
- Ensured robust application quality and stability by writing comprehensive unit and integration tests. Maintained high code quality standards through detailed code reviews and actively participated in Scrum ceremonies.

Strada — Senior Frontend Software Engineer 03/2023 – 10/2025

- Designed and developed applications within a microservices architecture to support efficient payroll management for global enterprise clients.
- Led incremental modernization of legacy modules by introducing **React 18** and **Next.js SSR** micro-frontends alongside Angular components to improve initial load performance (Core Web Vitals) and system adaptability.
- Implemented secure authorization mechanisms using Identity service and JSON Web Tokens (JWT) to ensure role-based access control across the platform.
- Built dynamic, reusable React hooks and shared components using **Zustand** and **TanStack Query** alongside Angular standalone components and Signals.
- Engineered and customized reusable web components, modules, and services to meet specific business requirements, enhancing system flexibility.
- Upgraded Angular applications by integrating Signals, Signal Store, and Standalone components for improved performance and maintainability.
- Utilized LLM workflows and automated AI code audits to reduce technical debt during Angular-to-React component refactoring.
- Maintained, updated, and optimized application architecture to ensure scalability, reliability, and consistent system performance.
- Contributed to project workflows by reviewing pull requests, partially configuring Azure DevOps pipelines, creating releases and tags, and supporting deployment processes.

SwissRe — Senior Frontend Software Engineer 09/2022 – 12/2022

- Developed and maintained map functionalities using **MapBox**, including tools for natural disaster analysis, drawing tools, custom map styles, location sets upload, and export features in Excel and PDF formats.
- Implemented hybrid component wrappers supporting both **React** and **Angular** dynamic map widgets for cross-team integration.
- Designed and updated Angular components and NgRx entities while leveraging **NX libraries** to create shared functionality across multi-team repositories.
- Supported and improved application architecture with strategic adjustments to optimize performance, achieving UI speed improvements of over **2x**.
- Conducted PR reviews, participated in Scrum meetings, sprint planning sessions, and managed release processes including creation of tags and releases in Azure DevOps.
- Wrote unit tests, technical specifications based on client communication, mentored junior developers, and integrated **Stripe** and **Google Play Billing** payment systems for seamless transactions.
- Created custom views for enhanced user interaction, resolved technical issues during app publication processes, and implemented new functionalities to meet project requirements effectively.

Alight — Frontend Software Engineer 03/2021 – 09/2022

- Led the development and management of large Angular and **Next.js SSR** forms, incorporating real-time data visualization through Charts and utilizing **SignalR** for dynamic data updates.
- Implementation of Server-Side Rendering (SSR) to enhance application performance, initial load speed, and SEO capabilities for financial modules.
- Delivery of modular Angular and React components, modules, and services aligned with project requirements, improving overall application functionality.
- Maintenance and architectural modifications to optimize performance and ensure adaptability in response to evolving project needs.
- Code review processes via pull requests (PRs) to maintain high-quality standards, active participation in Scrum meetings, and contribution to collaborative teamwork flows.
- Partial configuration of Azure DevOps pipelines, release management including tag creation, and customization of web components to streamline deployment processes.

NGA HR — Frontend Software Engineer 11/2020 – 03/2021

- Developed and implemented enterprise web components and modules using **React** and **Angular** to meet complex HR system requirements.
- Maintenance and enhancement of application architecture to ensure scalability and reliability across enterprise environments.
- Review of pull requests (PRs) for quality assurance and adherence to software development standards.
- Writing unit tests using **Jest** and **React Testing Library** / Jasmine to maintain code quality and prevent regression issues across applications.
- Contribution to recruitment processes by assisting in candidate interviews for technical roles.
- Active participation in Scrum ceremonies, including sprint planning and stand-ups, to ensure effective team collaboration and agile practices.
- Configuration and management of Azure DevOps pipelines, including creation of releases, tags, and partial pipeline setup.

MyCredit — Frontend Software Engineer 11/2018 – 11/2020

- Developed and maintained an admin tool using **React (Redux Toolkit)**, **Angular 5–8**, Angular Material, reactive forms, RxJS, NgRx, .NET back-end, and various third-party libraries to improve user experience and functionality.
- Processed and visualized business logic data for marketing purposes by integrating back-end REST APIs and third-party APIs such as Facebook API.
- Designed technical specifications based on client requirements through effective communication to ensure project alignment with business goals.
- Integrated external libraries and APIs to enhance application capabilities and streamline operations for better performance.
- Provided mentorship to junior developers, fostering skills development and team competency.
- Worked collaboratively with cross-functional teams to implement efficient solutions aligned with organizational objectives.

EDUCATION

Specialist's Degree in Computer Engineering
National Aerospace University – Kharkiv Aviation Institute
Faculty of Radio Systems for Aircraft · 01/2008 – 01/2013

LANGUAGES

English: Upper-Intermediate (B2) **Ukrainian:** Native